

PULSE System

Population Update and Linking for Sustainable Empowerment

"Empowering Communities through Population Insights."

Project Brief & System Requirements

Office of the Provincial Population Officer — Province of Cavite

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1. Background

The Office of the Provincial Population Officer (OPPO) in Cavite currently documents Responsible Parenthood and Family Planning (RPFP) activities — profiling sessions and RPFP seminars — using paper forms. Each activity generates an RPFP Form 1 (individual/couple profile) and a Documentation Report, which are then manually consolidated by staff into a monthly summary spreadsheet (e.g., "RPFP/Profiling 2025 – March").

This manual process is time-consuming and prone to error, and the office has identified that it is short-staffed (limited manpower) to keep up with data entry and report generation demands across four districts and dozens of barangays each month.

To address this, the office proposed building a digital system — internally developed with the help of OJT (on-the-job training) students with programming ability — to automate documentation, reduce duplicate data entry, and generate reports on demand.

2. Project Overview

2.1 Project Name

PULSE System — Population Update and Linking for Sustainable Empowerment

2.2 Tagline

"Monitoring People, Empowering Communities."

Alternative taglines considered:

- "Data for People, Development for All."
- "Connecting Population Data to Community Progress."
- "Empowering Communities through Population Insights." (selected)

2.3 Purpose

PULSE is an integrated monitoring and information system designed to track population data and support evidence-based planning for community development. It will help the OPPO and its field teams identify population trends, strengthen family planning programs, and promote development through timely, accurate, and centralized information — replacing manual spreadsheet consolidation with an automated data pipeline.

3. Problem Statement

- Manpower is limited; data encoding and report consolidation consume significant staff time.
- Monthly summary tables (barangay, dates, participant counts, age brackets) are built by hand from paper Documentation Reports.
- No systematic way to detect duplicate participant or activity entries across submissions.
- Report generation (by sex, age bracket, activity type, district, etc.) is manual and inflexible.
- Distinct data types — individual profiling (RPFP Form 1) versus group seminars/outreach — are not clearly separated in a structured database.

4. Current Workflow (As-Is)

- Field teams (District I–IV) conduct profiling sessions or RPFP seminars in barangays.

- Each activity is documented on paper: RPFP Form 1 (per participant/couple) and/or a Documentation Report (per activity/session).
- Forms are submitted to the Technical Services Section (Planning/Monitoring & Evaluation, and IEC Development).
- Staff manually consolidate submissions into a monthly tracking spreadsheet, organized by code, date, barangay, municipality, district, sex, and age bracket.
- Reports are reviewed and approved by the Provincial Population Officer (Eloisa L. Comia) and relevant section heads.

5. Proposed Workflow (To-Be, with PULSE)

- Field staff register an activity/session in PULSE (profiling or RPFP seminar), selecting municipality/city — barangay and district auto-populate.
- Participant data is encoded directly into the system (individual profile fields mirroring RPFP Form 1, or aggregate counts for seminars).
- The system checks incoming records against existing entries to flag potential duplicates before saving.
- Consolidated and filtered reports (by sex, age bracket, barangay, district, activity type, date range) are generated on demand instead of built manually.
- Documentation Report data (topics discussed, speakers, referrals, ratings) is linked to the corresponding activity record.
- Reports route through the same review/approval chain (Field Operations → Technical Services review → OPPO approval) but within the system rather than on paper.

6. Stakeholders

Role	Name(s)	Responsibility in PULSE
Provincial Population Officer (OIC)	Eloisa L. Comia	Final approval of reports and system outputs
Field Operation Section Head	Marieta B. San Juan	Oversees District I–IV data collection; approves field data
Technical Services Section Head	Analiza N. Gumapac	Reviews consolidated data and reports
MIS / Data Banking Team	Ariel Jarin, Arlene Salazar, Nicole Rafols	System data banking, encoding accuracy, population profile maintenance
Planning, Monitoring & Evaluation Team	Paul Mark Feranil, Sherrilyn Yu	Report preparation, plans/programs monitoring, technical support
IEC Development Team	Emerie Kris Nepomuceno, Clarissa Matienzo, Vina Mariz Perido, Nilo Bucao	Documentation report preparation, IEC materials, information dissemination
District Program Officers/Workers	District I–IV teams	Frontline data collection and encoding from field activities
Development Team	OJT student developers	System design, build, and maintenance under office supervision

7. System Requirements

7.1 Functional Requirements

ID	Requirement	Description
FR-01	Activity Registration	Allow staff to create a new activity record specifying type (Profiling or RPFP Seminar), date conducted, municipality/city, and barangay.
FR-02	Auto District/Barangay Mapping	Automatically determine District (I–IV) and available barangays once a municipality/city is selected, removing manual district entry.
FR-03	Individual Profiling Entry	Digitize RPFP Form 1 fields: participant/couple name, sex, civil status, birthdate/age, address/household ID, education, number of children, FP method used, traditional/modern FP status, signature capture or confirmation.
FR-04	Seminar / Outreach Entry	Capture Documentation Report fields: participants' profile (target vs. actual, by sex and age bracket), number of speakers and topics discussed, activities undertaken (registration, opening remarks, seminar proper, counseling, closing remarks, evaluation, referrals), remarks, and persons responsible.
FR-05	Duplicate Detection	Check new participant/activity entries against existing records (e.g., by name, address/household ID, or activity code + date + barangay) and flag likely duplicates before final save.
FR-06	Consolidated Reporting	Generate monthly/quarterly summary reports mirroring the existing 'RPFP/Profiling' tracking table: code, date, type, district, barangay, municipality, sex counts, age bracket breakdown, and documenter.
FR-07	Filtered / Ad-hoc Reports	Allow filtering and report generation by sex, age bracket, barangay, municipality, district, activity type, or date range.
FR-08	Referral Tracking	Record and report on referrals made during activities, linked to the source activity.
FR-09	User Roles & Routing	Support role-based access mirroring the office hierarchy (field encoder → Technical Services reviewer → OPPO approver), with basic approval status tracking.
FR-10	Data Export	Export consolidated and filtered reports (e.g., to spreadsheet/PDF) for offline submission or printing where still required.

7.2 Non-Functional Requirements

ID	Requirement	Description
NFR-01	Usability	Simple data-entry interface usable by field staff with varying technical skill, given limited manpower and training time.
NFR-02	Data Integrity	Prevent duplicate and inconsistent records; enforce required fields matching official form codes (civil status, education, FP method codes, etc.).
NFR-03	Performance	Report generation should return results quickly enough for use during live review/approval meetings.
NFR-04	Data Privacy	Handle personal participant data (names, addresses, household IDs) in line with

ID	Requirement	Description
		the RFPF Form's data privacy disclaimer and applicable privacy regulations.
NFR-05	Maintainability	System should be maintainable by OJT developers with ongoing office supervision, given limited in-house technical staff.
NFR-06	Auditability	Retain a record of who encoded or edited each entry and when, to support review and approval workflows.

8. Data Sources / Data Model Reference

PULSE's data model should be informed by three existing paper artifacts currently used by the office:

8.1 RPPF Form 1 (POPCOM, Form RP1-PMED-FM021)

Individual/couple-level profiling form capturing: name, sex, civil status, birthdate/age, address/household ID, highest educational attainment, number of children, modern FP method used, intention to shift methods, traditional FP type/status, reason for using/intending FP, and participant signature.

8.2 Documentation Report (Office of the Provincial Population Officer)

Per-activity report capturing: city/municipality, barangay, venue, seminar type and date conducted, participants' profile (target vs. actual by sex and age bracket: 10–14, 15–19, 20 and above), number and names of speakers, topics discussed, activities undertaken, remarks, and documented-by / reviewed-by / approved-by sign-off chain.

8.3 Monthly Consolidated Tracking Table

Currently a manually built spreadsheet (e.g., 'RPPF/Profiling 2025 – March') with columns: code, date of conduct, type of conduct (profiling / RPPF seminar), district, barangay, municipality/city, sex counts, total, age bracket breakdown, and documented-by. This is the primary report PULSE should be able to auto-generate.

9. Success Criteria

- Elimination of manual re-typing of data from paper forms into a spreadsheet for monthly reporting.
- Zero or near-zero duplicate participant/activity records in consolidated reports.
- Ability to generate a filtered report (e.g., by district, barangay, or age bracket) in a few clicks rather than by manual recompilation.
- Reduced turnaround time between field data collection and report approval.
- Positive feedback from Field Operation and Technical Services staff on ease of encoding.

10. Open Questions / Items to Clarify

- Exact list of documents required to be gathered/submitted before an activity record is considered complete.
- Precise duplicate-matching logic (which fields determine a 'duplicate': name, household ID, or activity code + date + barangay combination).
- Whether ratings/evaluation data (e.g., speaker ratings) should be part of the same system or a separate module.
- Data retention and privacy handling requirements for personally identifiable participant information.
- Final decision on system name/branding: PULSE System vs. PULSE+.